Advanced material Physics

Duy M. Dao

October 27, 2026

Part I

Dielectric Polarization under Static Electric Field

1 Lorentz Local field

Under static external electric field, a single molecule within a material will feel a local electric field $\vec{E}_{loc}$ 4 different components: the external field itself $\vec{E}_0$,the depolarization field caused by the bound charge $\vec{E}_1$, and the field caused by local neighbor dipoles/molecules.

For an uniform, homogenous material, we can approximate the field felt by a molecule by calculating only the field within a certain shape that is big enough so that the discrete molecules can be treated as a continuum and the field of the molecules outside it is neglegible; and small enough so that the polarization within it can be considered constant. The last compononent can be decomposed into the dipole contribution $\vec{E}_3$ and the bound charge contribution on the surface of that shape $\vec{E}_4$.

The shape of choosing intuitively should be a sphere although every shape should obtain the same answer. A sphere covers all lattice symmetry therefore work well with most lattice.

We choose a convenient coordinate system to deal with this system: the $\hat{z}$ axis is in the same direction with the polarization vector P there. The center of the sphere is right at our molecules

The dipole moment within the sphere, by symmmetry should be 0. since a dipole electric field is:

$$\vec{E}(\vec{R}) = \frac{3(\vec{p}\cdot\hat{R}) - \hat{p}}{R^3} \tag{1.1}$$